

Supplementary Information

Manipulating charge transfer dynamics and stabilizing lead bromide octahedra for efficient blue perovskite light-emitting diodes

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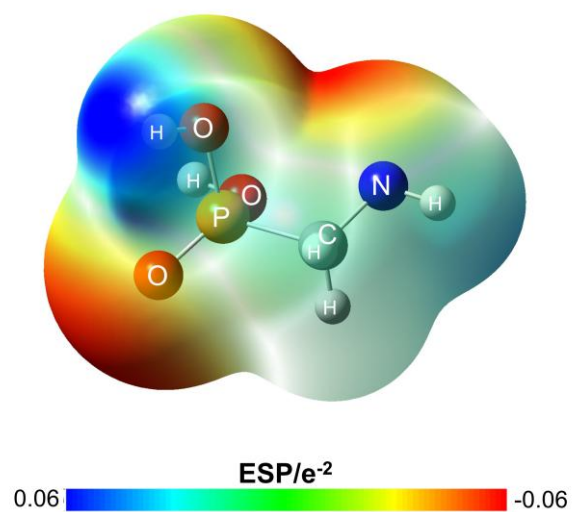
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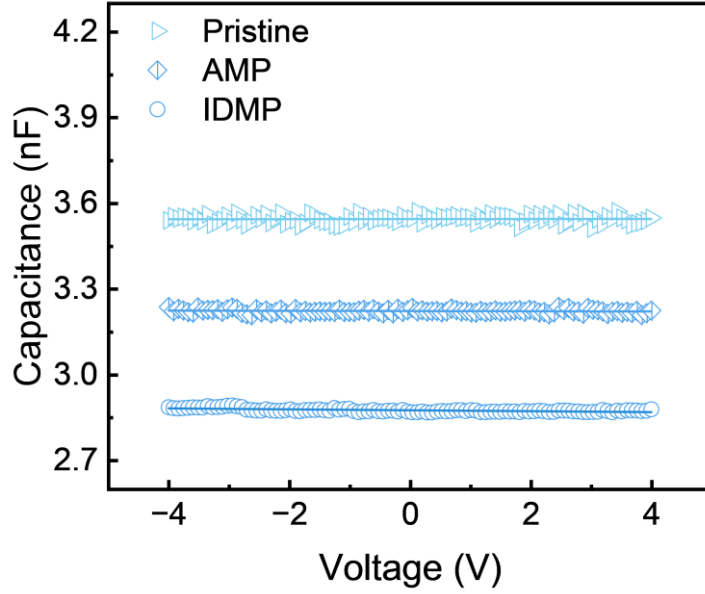
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Supplementary Figure 1 | ESP mapping of AMP molecules. Electrostatic potential (ESP) mapping of (Aminomethyl)phosphonic acid (AMP) molecular.

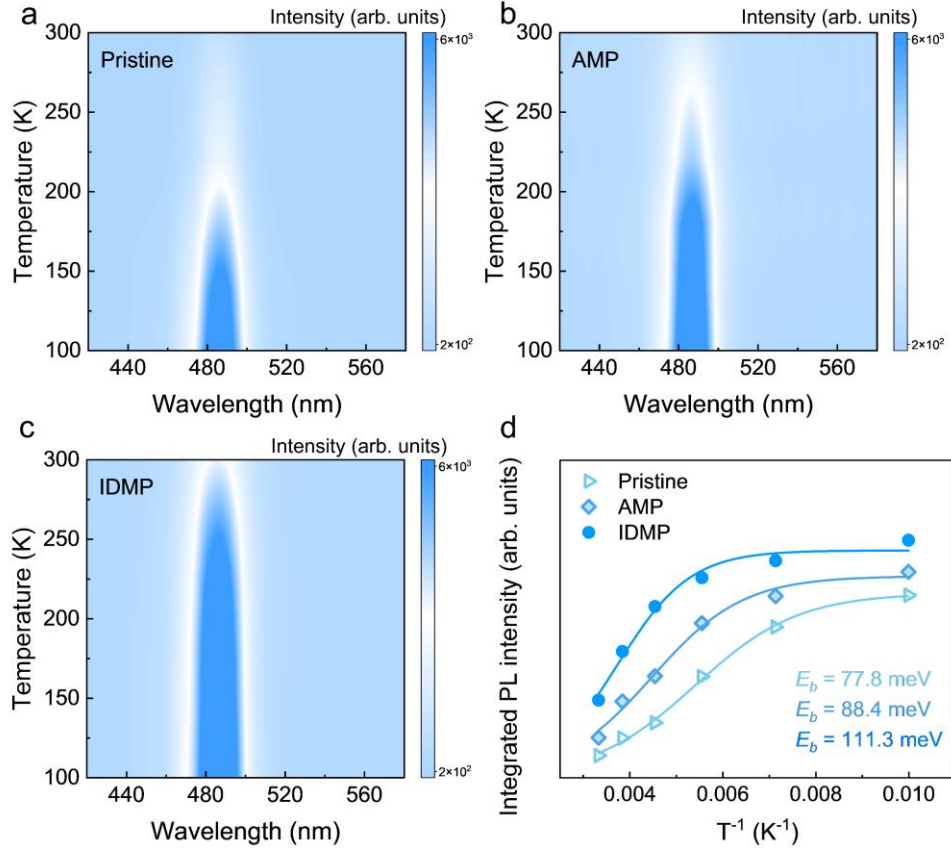


Supplementary Figure 2 | Capacitance-voltage measurements. Capacitance-voltage curves for pristine, AMP-treated, and IDMP-treated perovskite films.

Supplementary Discussion: We prepared the structure of ITO/PEDOT:PSS/PVK:PBD/2PACz/Perovskite/3TPYMB/LiF/Al device to test the capacitance of pristine, AMP-treated, and IDMP-treated perovskite films. Their relative dielectric constant (ϵ) values were calculated using the formula:

$$C = \epsilon \epsilon_0 S / D \quad (1)$$

where C is the capacitance of the dielectric, ϵ_0 is the dielectric constant of vacuum, S is the area and D is the thickness of the film. The S of all perovskite films are 4 mm^2 , and the D for all samples is 80 nm.

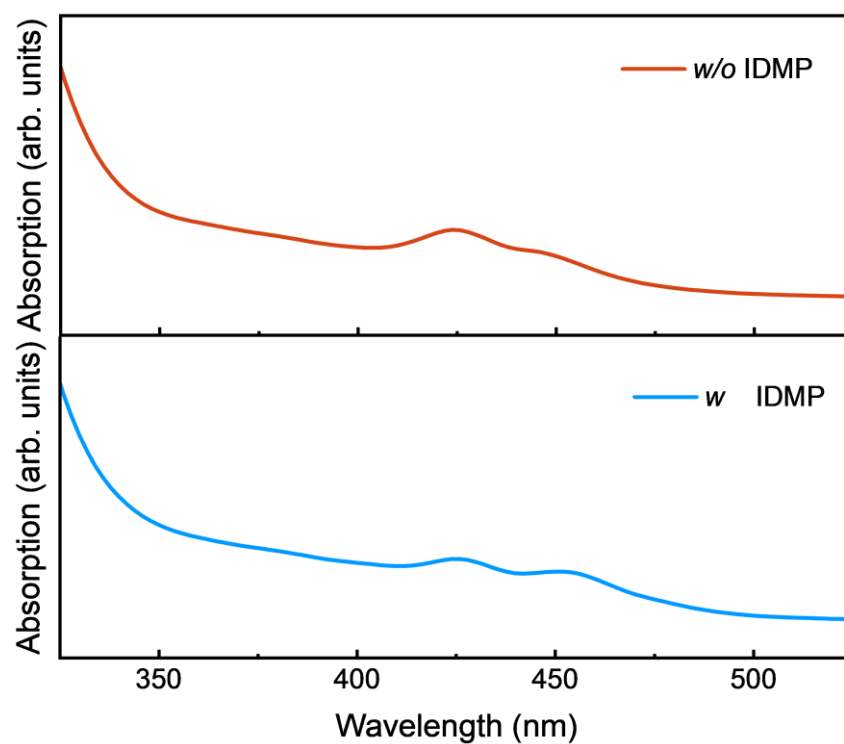


Supplementary Figure 3 | Temperature-dependent PL spectra of the perovskite films. 2D color maps of temperature-dependent PL spectra for (a) pristine, (b) AMP-treated, and (c) IDMP-treated perovskite films. (d) Integrated PL intensity and derived exciton binding energy of pristine, AMP-treated, and IDMP-treated perovskite films as a function of reciprocal temperature.

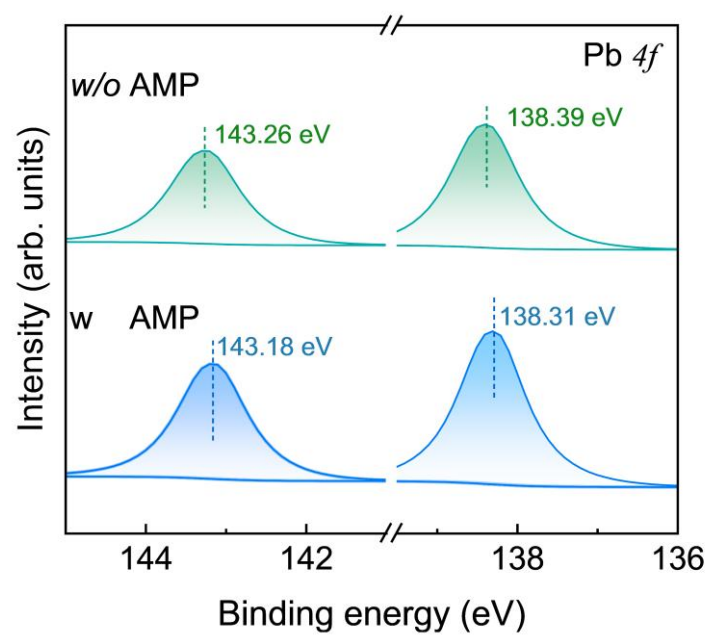
Supplementary Discussion: Temperature-dependent PL measurements were performed across a range of temperatures from 100 to 300 K (Supplementary Fig. 3a-c). The temperature-dependent PL spectra were excited by a 405 nm laser and PL intensity was plotted as a function of temperature, as shown in Supplementary Fig. 3d. The E_b can be calculated using the Arrhenius equation:

$$I(T) = I_0 / [1 + A \exp(-E_b/k_B T)] \quad (2)$$

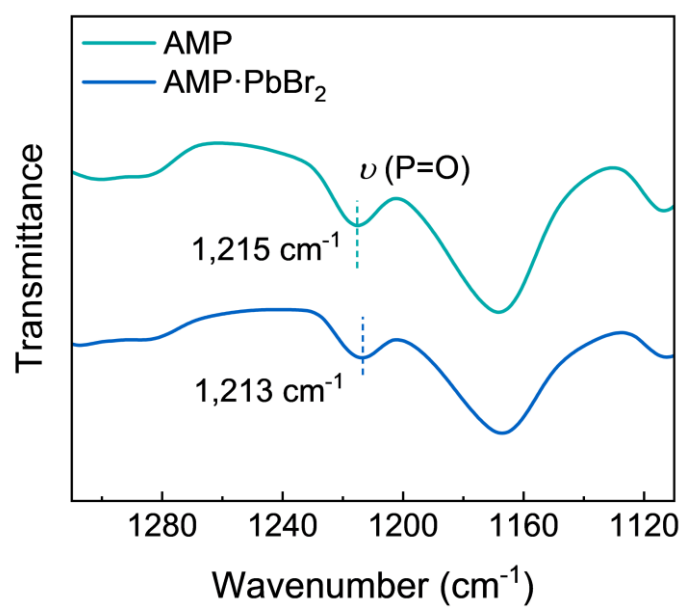
where $I(T)$, I_0 , k_B and A represent the PL intensity at temperature T , the PL intensity at initial temperature, Boltzmann constant, and a fitting parameter, respectively. Specifically, pristine films demonstrate an E_b of 77.8 meV, whereas films subjected to AMP and IDMP treatment display a larger E_b of 88.4 meV and 111.3 meV, respectively.



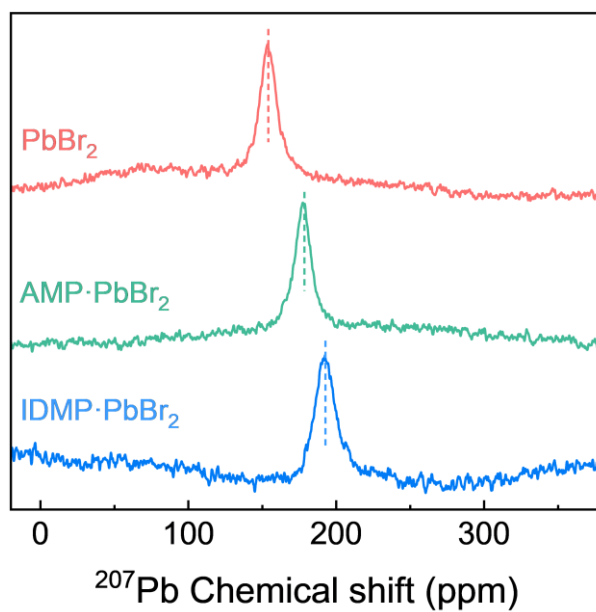
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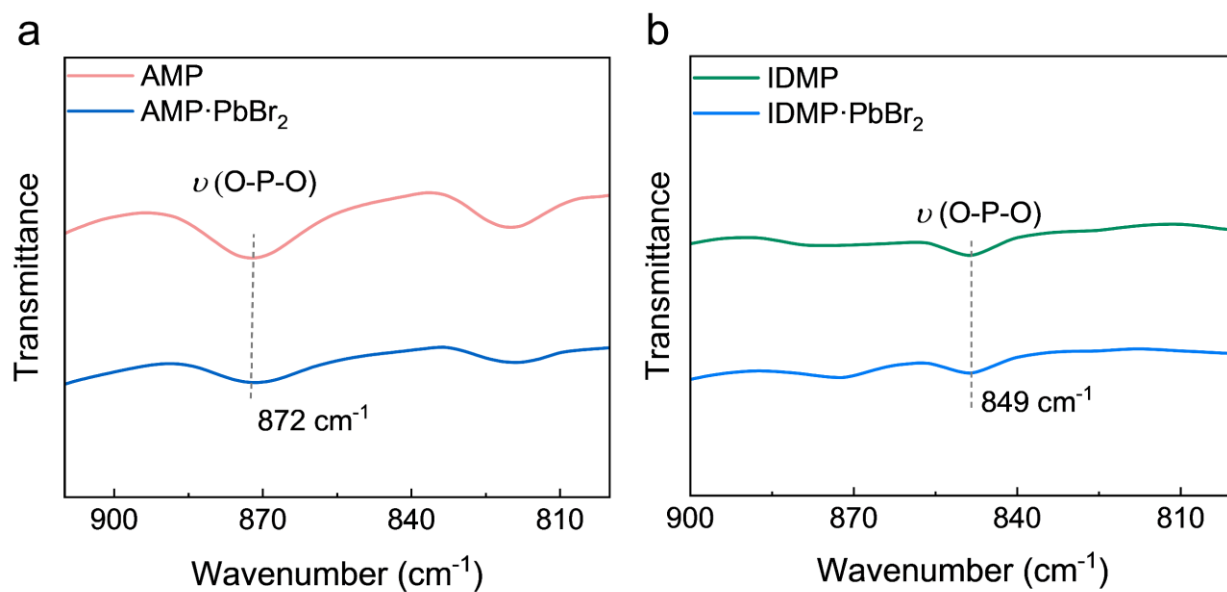
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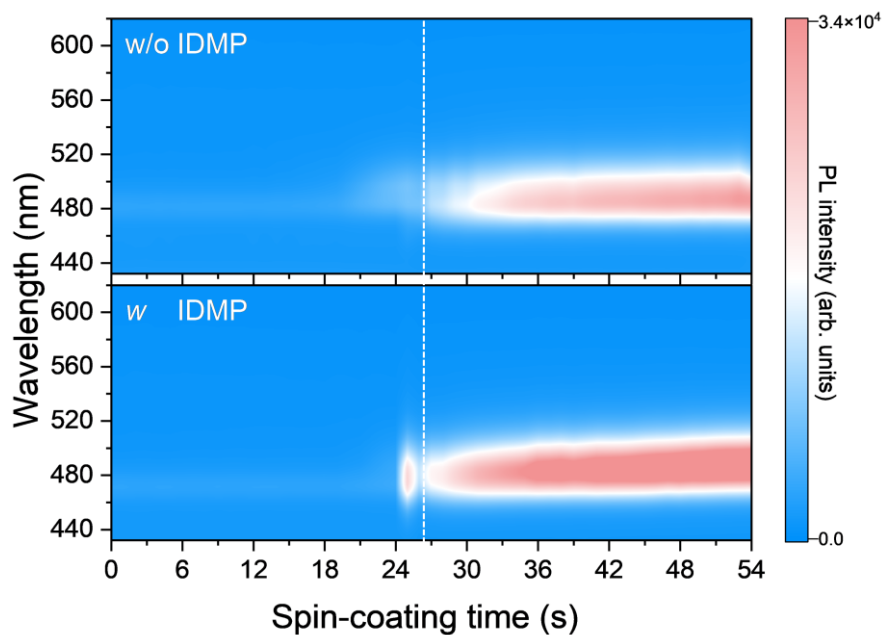
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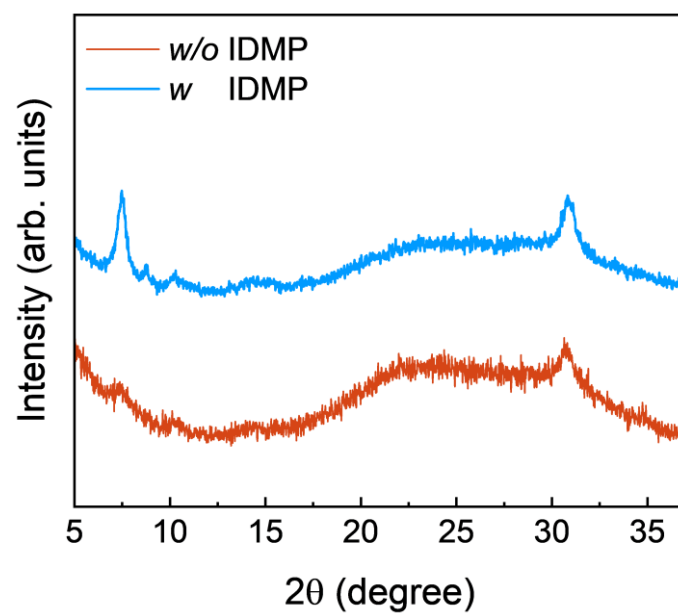
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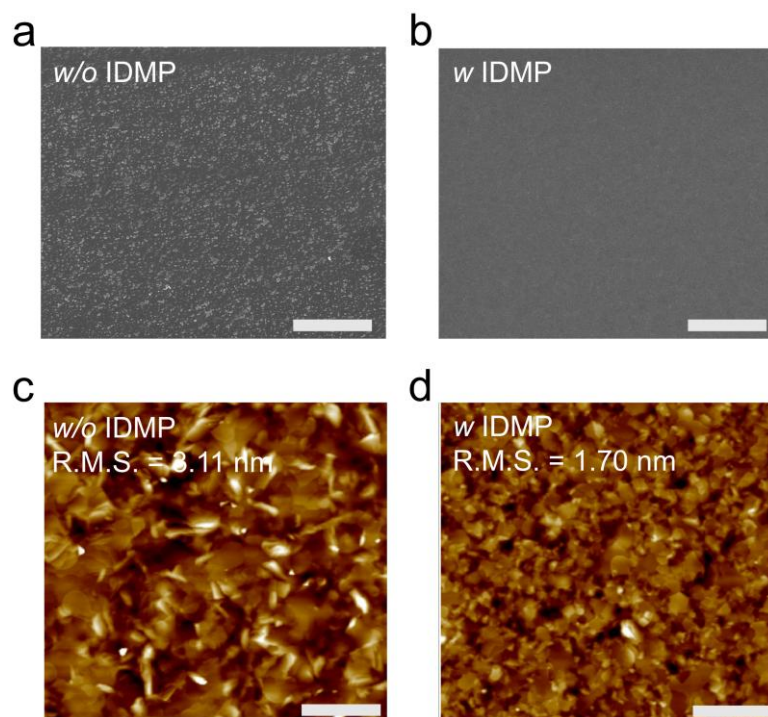
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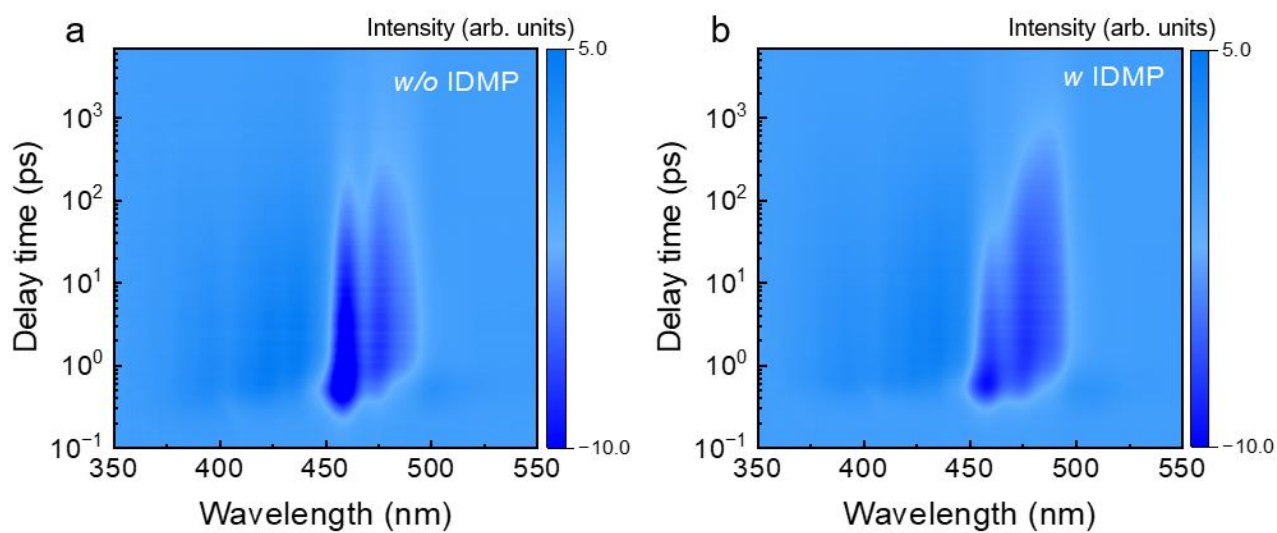
Supplementary Figure 11 | Time-resolved in situ PL spectra of perovskite films. Time-resolved in situ PL spectra during the spin-coating process of the pristine and IDMP-treated perovskite films. The perovskite precursors were spin-coated at 3000 rpm for 54 s. At 27 s of spin-coating, the antisolvent ethyl acetate was dropped to promote the crystallization, and then annealed at 60 °C for 5 min.



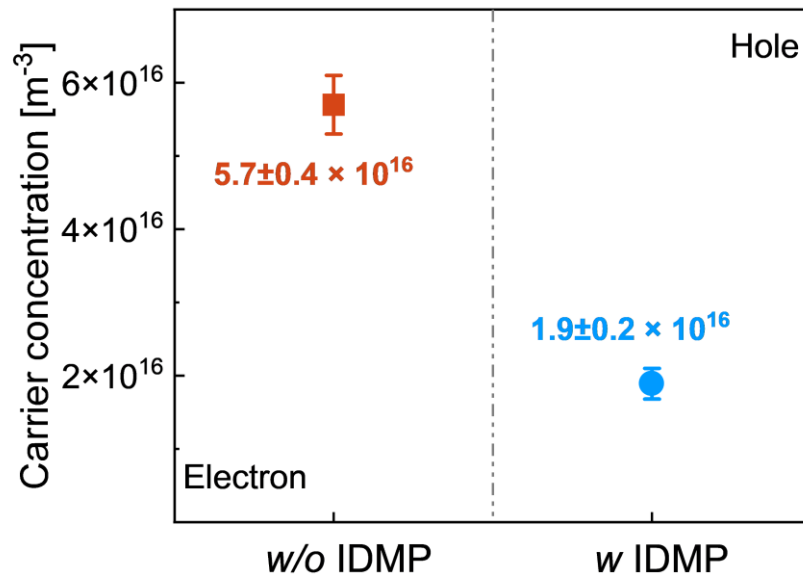
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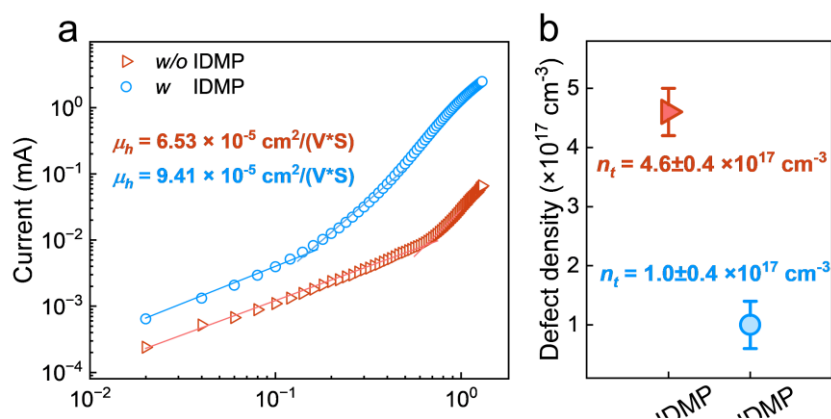
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Supplementary Figure 14 | 2D pseudo-color contour mapping of perovskite films. 2D pseudo-color contour mapping for **a** pristine, and **b** IDMP-treated perovskite films.

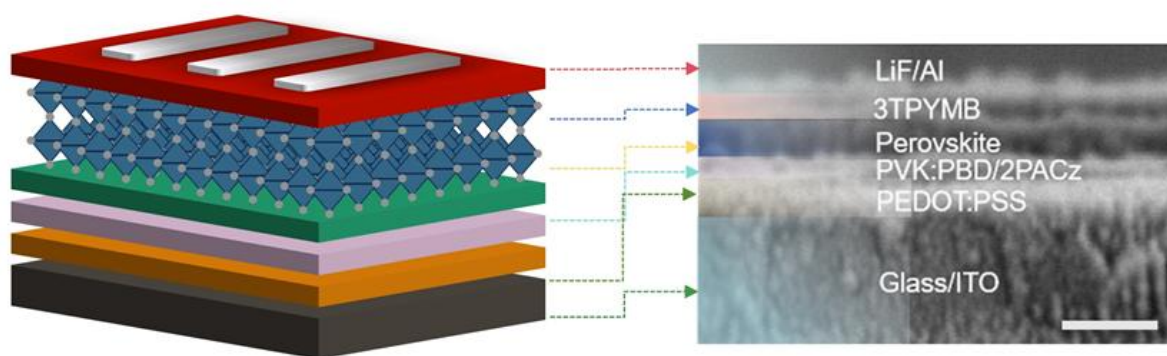


Supplementary Figure 15 | Carrier concentration measurements of perovskite films. Carrier concentration as a function of doping level. The dots represent the average values. The error bars mark the upper and lower bounds of the 95% confidence interval.

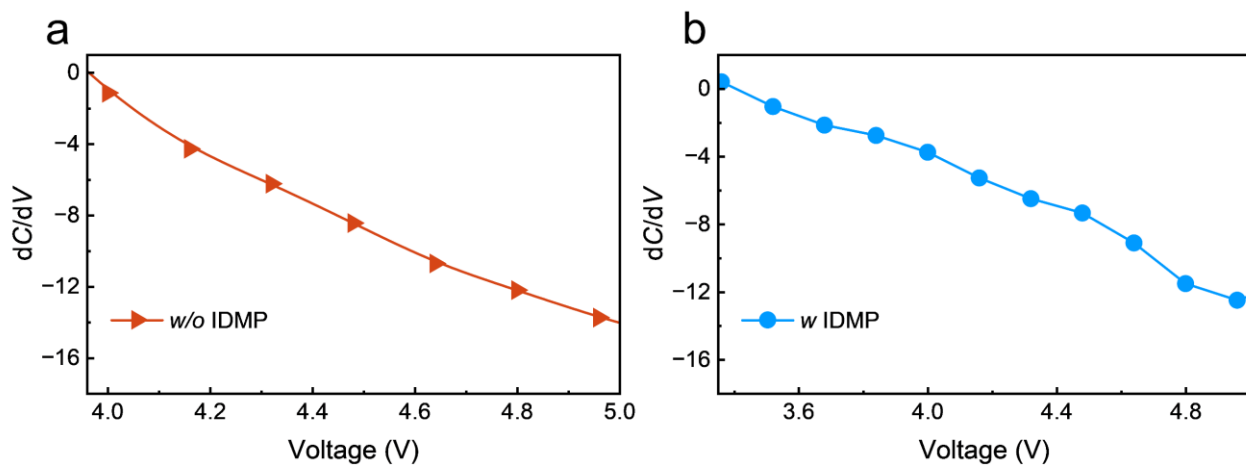


Supplementary Figure 16 | SCLC measurements an

current measurements of hole-only devices based on pristine and IDMP-treated perovskite films. The structure of hole-only devices was ITO/PEDOT:PSS/PVK:PBD/2PACz/perovskites/MoO₃/Al. **b** The corresponding calculated defect state densities.



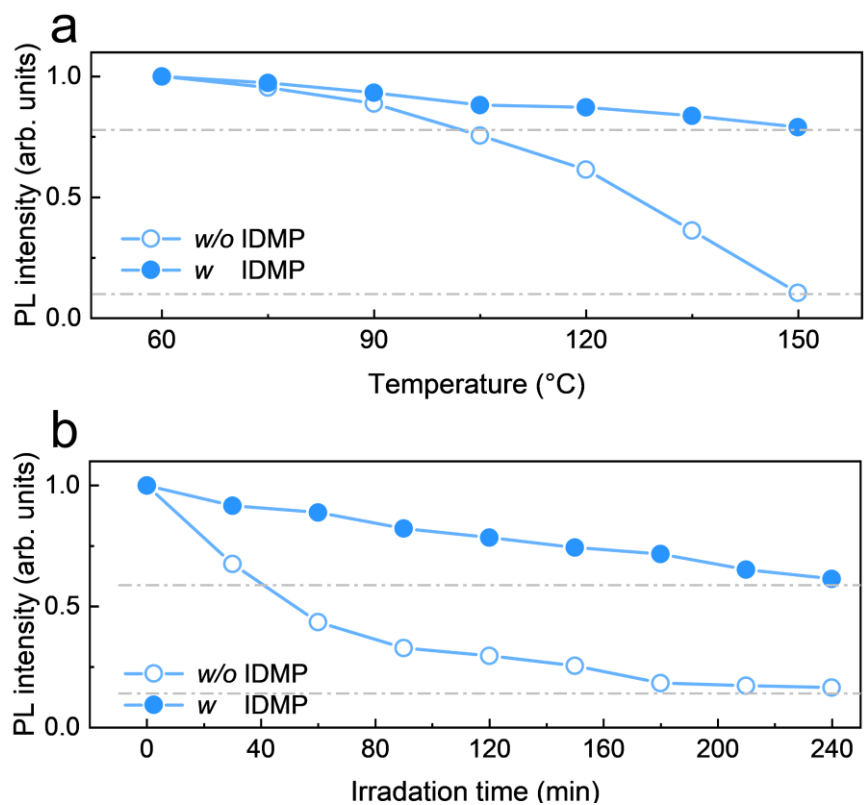
Supplementary Figure 17 | Structure of the device. The structure and cross-section SEM image of the device, scale bars, 200 nm.



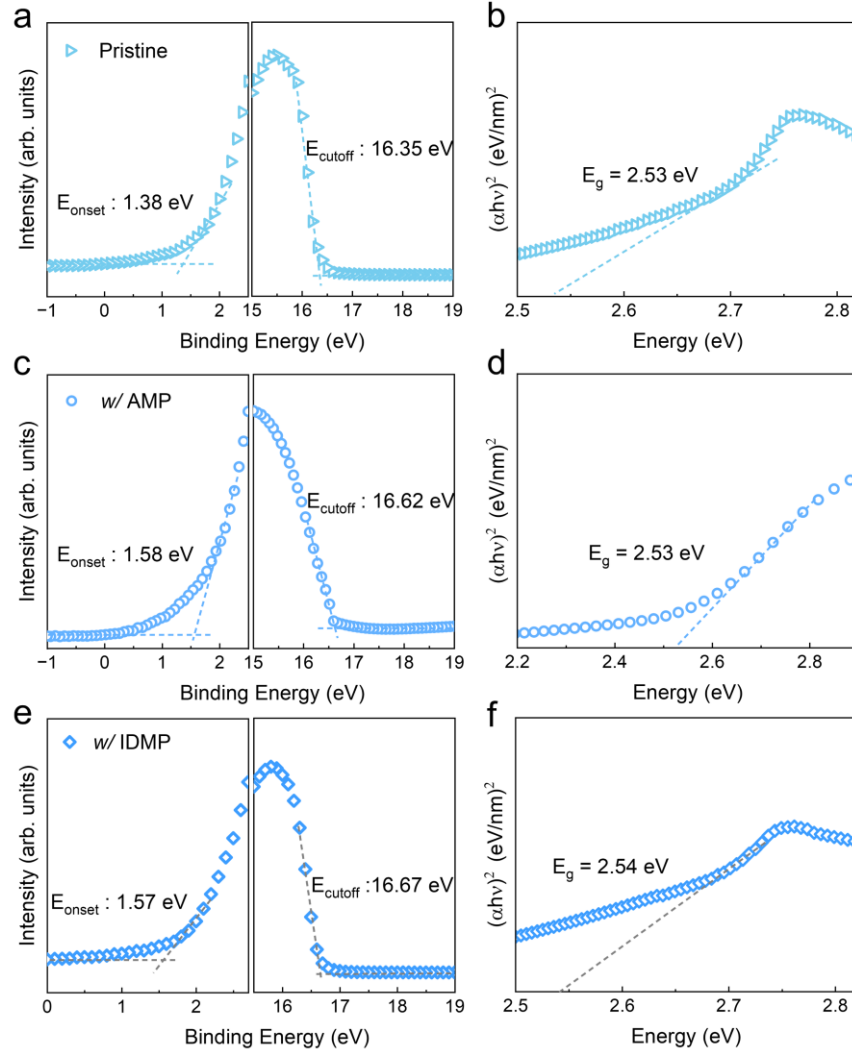
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Supplementary Figure 19 | Time-dependent microscopic images of perovskite films. Time-dependent microscopic images of AMP-treated perovskite films under an external electric field ($\sim 6 \times 10^4 \text{ V m}^{-1}$). Scale bars, $20 \text{ }\mu\text{m}$. The bilateral black areas refer to Ag electrodes and the external electric field is marked by the “+” and “-” symbols.

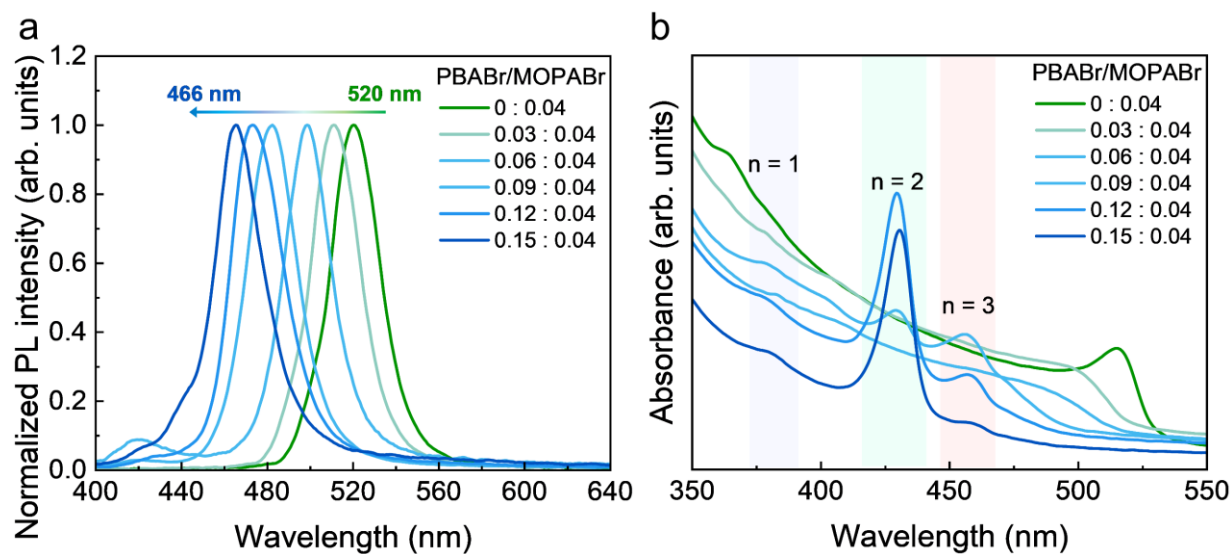


Supplementary Figure 20 | Steady PL measurements of perovskite films. Normalized PL spectra of pristine and IDMP-treated perovskite films under **a** heating and **b** irradiation conditions.

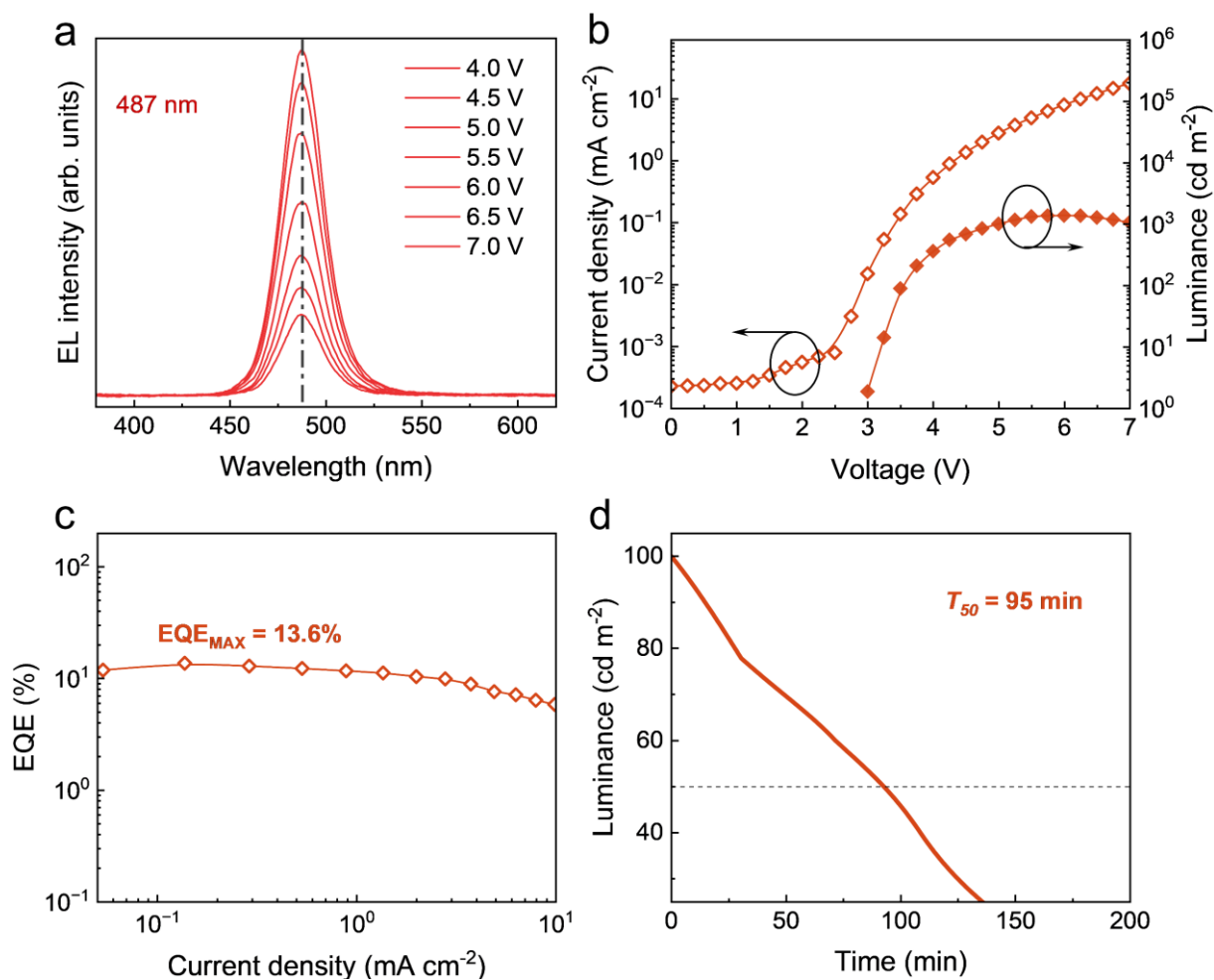


Supplementary Figure 21 | UPS measurements of perovskite films. UPS spectra of **a** the pristine, **c** AMP-treated, and **e** IDMP-treated perovskite films. Tauc plot analysis for **b** the pristine, **d** AMP-treated, and **f** IDMP-treated perovskite films.

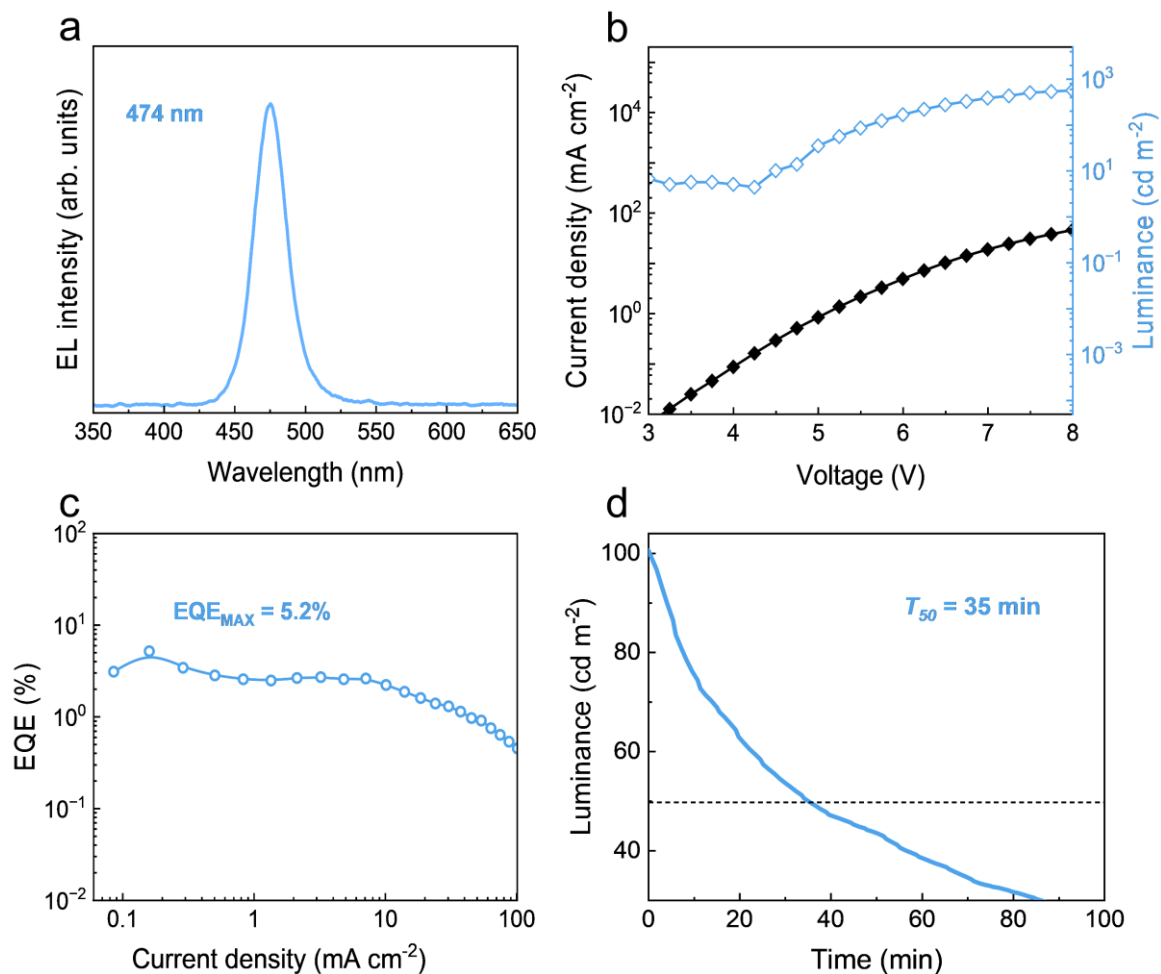
Supplementary Discussion: The energy levels of the perovskite layers were calculated from the UV/Vis absorption and ultraviolet photoelectron spectroscopy (UPS) spectra. UPS spectra are conducted to explore the influence of the perovskite film with AMP and IDMP. From the secondary electron cutoff (E_{cutoff}) region in following figures, the work functions ($WF = 21.22 \text{ eV} - E_{\text{cutoff}}$) of the pristine, AMP-treated, and IDMP-treated perovskite film are 4.87, 4.60, and 4.55 eV, respectively. The corresponding valence band maxima (VBM) is 6.25, 6.18, and 6.12 eV, respectively ($VBM = -(WF + E_{\text{onset}})$). And the conduction band minimum (CBM) is 3.72, 3.65, and 3.58 eV, respectively ($CBM = -(VBM - E_g)$).



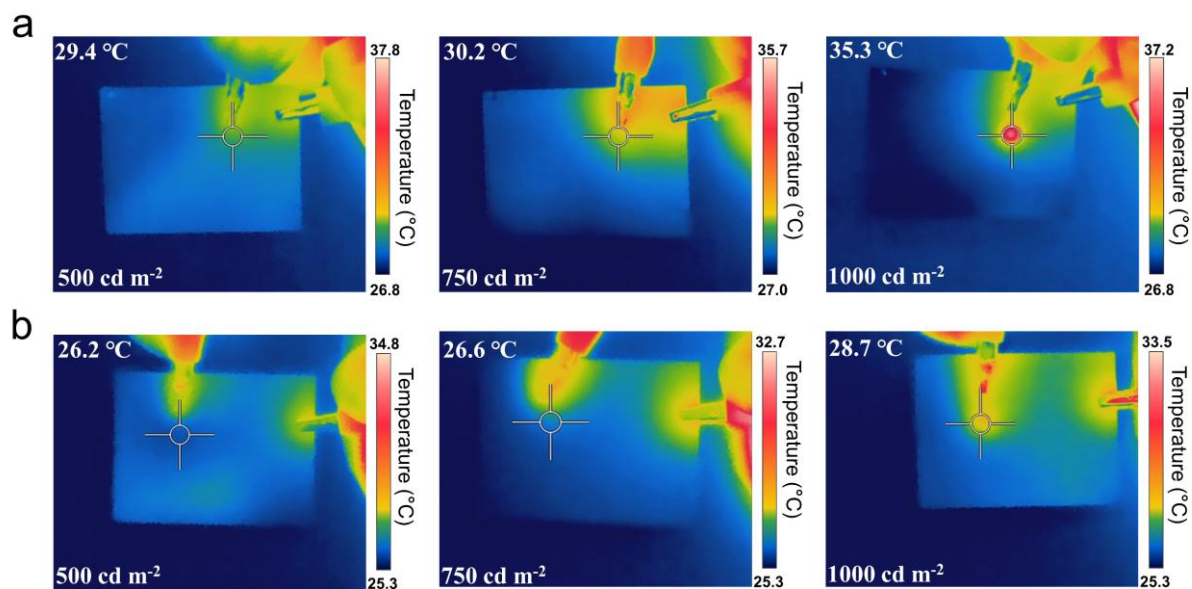
Supplementary Figure 22 | PL and UV-vis absorption measurements of perovskite film. **a** PL spectra and **b** UV-vis absorption of perovskite films based on PBABr/MOPABr with different molar ratios.



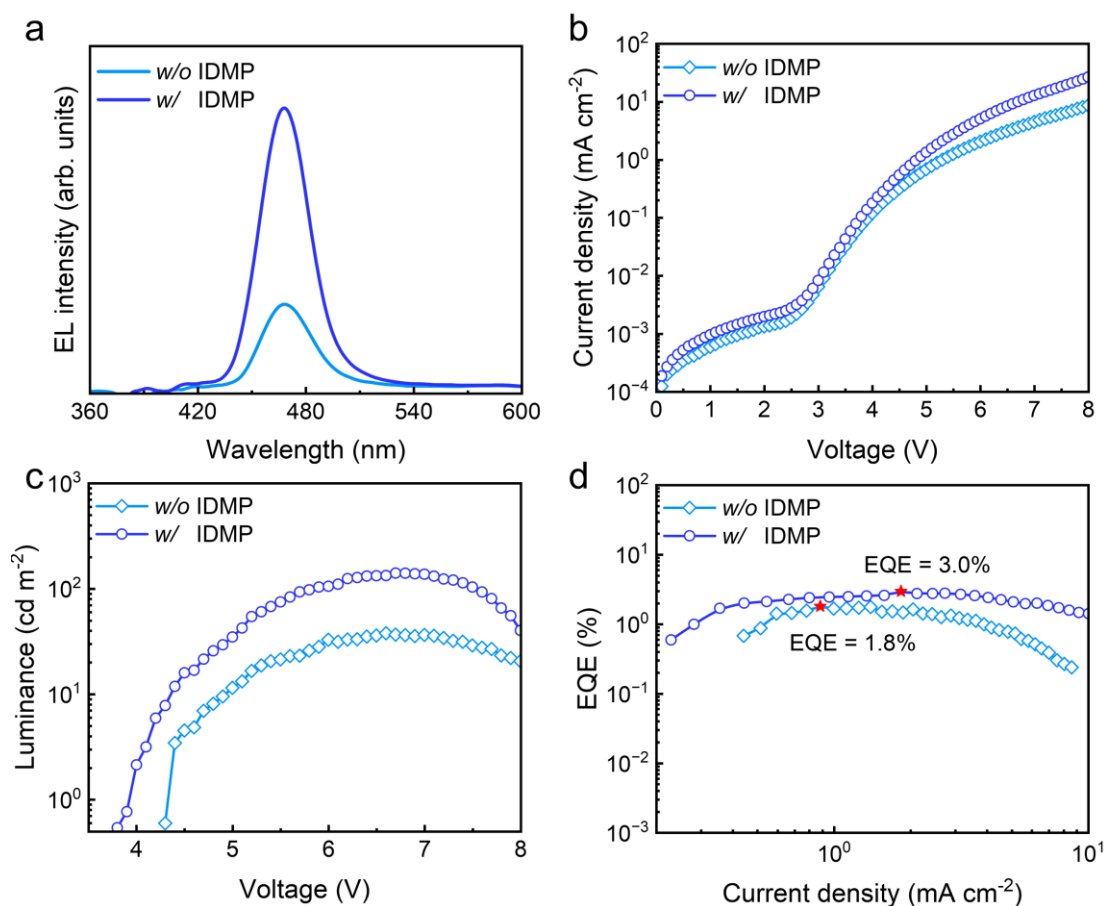
Supplementary Figure 23 | Device performance of control sky-blue PeLEDs. EL performance for control device (emitting at 487 nm). **a** EL spectra with a constant peak emission wavelength of 487 nm, **b** J - V - L , **c** EQE - J curves, and **d** operational lifetime. The champion control PeLED emitting at 487 nm shows an enhanced EQE up to 13.6%, and the operating half-lifetime (T_{50}) at an initial luminance level of 100 cd m^{-2} reached approximately 95 minutes.



Supplementary Figure 24 | Device performance of control pure-blue PeLEDs. EL performance for control device (emitting at 474 nm). **a** EL spectra with a constant peak emission wavelength of 474 nm, **b** J - V - L , **c** EQE - J curves, and **d** operational lifetime. The peak EQE for the champion control device emitting at 474 nm is 5.2%, and the operating half-lifetime (T_{50}) at an initial luminance level of 100 cd m^{-2} is approximately 35 minutes.



Supplementary Figure 25 | Surface temperature measurements of PeLEDs. Surface temperature of operating sky-blue PeLEDs based on **a** pristine and **b** IDMP-treated perovskite films.



Supplementary Figure 26 | Device performance of deep-blue PeLEDs. EL performance for deep-blue PeLEDs based on pristine and IDMP-treated perovskite films. **a** EL spectra peaking at 468 nm, **b** J - V , **c** L - V , and **d** EQE - J curves. The peak EQEs for the champion control and target devices emitting at 468 nm are 1.8% and 3.0%, respectively.

Supplementary Table 1 | PL lifetime data of perovskite films. The parameters of carrier lifetime by fitting the TRPL spectroscopy based on the pristine, AMP-treated, and IDMP-treated perovskite films.

	A_1	τ_1 (ns)	A_2	τ_2 (ns)	A_3	τ_3 (ns)	τ_{avg} (ns)
Pristine	0.42	3.42	0.46	0.79	0.12	17.68	10.89
AMP	0.45	3.59	0.36	0.92	0.19	19.21	13.62
IDMP	0.78	1.24	0.26	4.53	0.02	25.04	17.51

Supplementary Discussion:

The TRPL curves were fitted by a tri-exponential rate law:

$$\tau_{avg} = \frac{A_1\tau_1^2 + A_2\tau_2^2 + A_3\tau_3^2}{A_1\tau_1 + A_2\tau_2 + A_3\tau_3} \quad (3)$$

Where A_1 , A_2 , and A_3 are the decay amplitudes, the fast decay component τ_1 is related to the trap-assisted non-radiative recombination, and slow decay time constant τ_2 and τ_3 associates with the radiative recombination in the energy transfer process of the perovskite films.

Supplementary Table 2 | The parameters of recombination rate constants. The parameters of radiative and nonradioactive recombination rate constants by fitting the TRPL and PLQY values based on the pristine, AMP-treated, and IDMP-treated perovskite films.

	τ_{avg} (ns)	PLQY (%)	k_r (/s)	k_{nr} (/s)
Pristine	10.89	21	1.89×10^7	7.11×10^7
AMP	13.62	39	2.86×10^7	4.44×10^7
IDMP	17.51	70	3.99×10^7	1.72×10^7

Supplementary Discussion:

The recombination rate can be calculated from the PLQY and the average lifetime of the perovskite films. The average lifetime is determined by the radiative and non-radiative recombination rate constants (k_r and k_{nr}):

$$\tau_{avg} = 1/(k_r + k_{nr}) \quad (4)$$

The radiative can be calculated from the equation:

$$PLQY = k_r / (k_r + k_{nr}) \quad (5)$$

Supplementary Table 3 | TA fitting parameters of perovskite films. The fitting parameters of TA spectra based on without or with IDMP-treated perovskite films.

	Film	A ₁	τ ₁ (ps)	A ₂	τ ₂ (ps)	A ₃	τ ₃ (ps)	τ _{rise} (ps)
458 nm	w/o IDMP	-0.33	1.65	-0.43	27.4	-0.21	333	0.21
	w IDMP	-0.40	0.55	-0.27	7.69	-0.30	73	0.24
477 nm	w/o IDMP	-0.29	14.4	-0.4	145	-0.19	1360	0.45
	w IDMP	-0.20	13.4	-0.45	170	-0.25	1010	0.38

Supplementary Discussion:

The kinetics of each GSB can be fitted using a multiexponential function:

$$\Delta A(t) = a_1 \exp(-t/\tau_1) + a_2 \exp(-t/\tau_2) + a_3 \exp(-t/\tau_3) - c_1 \exp(-t/\tau_{rise}) \quad (6)$$

where a_1 , a_2 , a_3 , and c_1 represent amplitudes; τ_1 denotes the fast decay time constant associated with carrier transfer processes within the perovskite; τ_2 and τ_3 signify slow decay time constants; while τ_{rise} indicates formation time constant. The fitting parameters have been extracted and listed in Supplementary Table 3.

Supplementary Table 4 | Summarized device parameters of blue PeLEDs. Summary of performance for sky-blue, pure-blue and deep-blue LEDs based on pristine and IDMP-treated pure-Br perovskite films.

	PeLED	λ_{EL} (nm)	V_{on} (V)	Luminance (cd m^{-2})	EQE (%)	T_{50} (min)
sky-blue	w/o IDMP	487	3.0	1373	13.6	95
	w/ IDMP	487	3.0	2459	25.4	845
pure-blue	w/o IDMP	474	3.0	596	5.2	35
	w/ IDMP	474	3.0	826	16.3	392
deep-blue	w/o IDMP	468	3.8	38	1.8	N/A
	w/ IDMP	468	4.2	141	3.0	N/A

Supplementary Table 5 | Summary device parameters of pure-Br and mixed halide PeLEDs.

Summary of performance parameters of representative blue LEDs based on pure-Br and mixed halide perovskites reported in literatures.

	Perovskite	λ_{EL} (nm)	Luminance (cd m ⁻²)	EQE (%)	Stability (T ₅₀)	Ref
	CsPbBr ₃ nanoplatelets (NPLs)	455	1511	4.15	50 min @ 100 cd m ⁻²	1
	CsPbBr ₃ NPLs	460	591	1.60	N/A	2
	CsPbBr ₃ NPLs	462	691	1.77	20 min @ 100 cd m ⁻²	3
	CsPbBr ₃ NPLs	464	4.6	0.70	N/A	4
	CsPbBr ₃ NPLs	465	227	5.4	860 s @ 100 cd m ⁻²	5
deep-	PEA ₂ (Cs _{0.6} EA _{0.4} PbBr ₃) ₂ PbBr ₄	466	N/A	3.64	81 s @ 30 cd m ⁻²	6
blue	(CHDA _{1-x} CHMA _x) (Cs _{1-m} FA _m) _{n-1} Pb _n Br _{3n+1}	466	121	9.22	125 s @ 11 cd m ⁻²	7
	(p-FPEA) ₂ Cs _{1.5} Pb _{2.5} Br _{8.5}	469	2428	10.4	57 min @ 100 cd m ⁻²	8
	CsPb(Br _x /Cl _{1-x}) ₃ NCs	460	255	12.0	~75 s @ 1 mA cm ⁻²	9
	CsPb(Br/Cl) ₃	461	1978	5.68	4.1 min @ 100 cd m ⁻²	10
	PEA ₂ (Cs/FA) _{n-1} Pb _n (Br/Cl) _{3n+1}	467	448.81	4.42	78 s @ 100 cd m ⁻²	11
	CsPb(Br/Cl) ₃	468	282	16.3	N/A	12
	p-F-PEABr ₂ Cs _{n-1} Pb _n (Br/Cl) _{3n+1}	468	~100	15	253 s @ 1 mA cm ⁻²	13
	CsPb(Br/Cl) ₃ QDs	469	31407	20.3	T ₈₀ 4.8 h @ 100 cd m ⁻²	14
	PBA ₂ (Cs _{1-m} FA _m PbBr ₃) _{n-1} PbBr ₄	474	~ 200	4	N/A	15
	PEA ₂ (CsPbBr ₃) _{n-1} PbBr ₄	478	512	5.52	80 s @ 99 cd m ⁻²	16
	CsPb(Br _x /Cl _{1-x}) ₃ NCs	470	635	21.3	~80 s @ 1 mA cm ⁻²	9
	CsPbCl _{3-x} Br _x QDs	470	1176	24.9	77 min @ 100 cd m ⁻²	17
pure-	o-F-PEA ₂ (GA/Cs) _{n-1} Pb _n (Br _x /Cl _{1-x}) _{3n+1}	472	1297	9.32	15 min @ 100 cd m ⁻²	18
blue	(p-F-PEA/PBA) ₂ (Li/GA/Cs)Pb(Br/Cl) ₃	472	267.5	15.1	T ₇₀ : 16 min @ 100 cd m ⁻²	19
	(PEA/PPA/EDAD) ₂ Cs _{n-1} Pb _n (Br _x /Cl _{1-x}) _{3n+1}	476	943	16.61	N/A	20
	(FA/Rb/Cs)Pb(Br _x /Cl _{1-x}) ₃	478	8136	10.6	N/A	21
	(MOPA/PBA)₂Cs_{n-1}Pb_nBr_{3n+1}	474	826	16.3	392 min @ 100 cd m⁻²	Our work
sky-	CsPbBr ₃ quantum dots (QDs)	480	~ 2000	18	2 h @ 100 cd m ⁻²	22
blue	PBA ₂ (FA/Cs) _{n-1} Pb _n Br _{3n+1}	483	~ 700	9.5	250 s @ 100 cd m ⁻²	23

$(\text{PA/PEA})_2\text{CsPbBr}_3$	486	1061	16.2	600 s @ 0.1 mA cm ⁻²	24
$\text{PEA}_2(\text{Cs}_{1-x}\text{EA}_x\text{PbBr}_3)_2\text{PbBr}_4$	488	2191	12.1	1 h @ 100 cd m ⁻²	25
$(\text{BDA}_{1-x}\text{PEA}_{2x})(\text{CsPbBr}_3)_{y-1}\text{PbBr}_4$	489	716	13.7	154 s @ 100 cd m ⁻²	26
$\text{DADOCs}_{n-1}\text{Pb}_n\text{Br}_{3n+1}$	493	3028	16.55	52 s @ 100 cd m ⁻²	27
$\text{BEA}_2\text{Cs}_{n-1}\text{Pb}_n\text{Br}_{3n+1}$	493	2540	16.98	730 s @ 0.5 mA cm ⁻²	28
$\text{CsPb}(\text{Br}_x/\text{Cl}_{1-x})_3 \text{ NCs}$	480	1601	26.4	~100 s @ 1 mA cm ⁻²	9
$(\text{PEA/PPA/EDAD})_2\text{Cs}_{n-1}\text{Pb}_n(\text{Br}_x/\text{Cl}_{1-x})_{3n+1}$	485	4384	24.03	160 min @ 100 cd m ⁻²	20
$(\text{FA/Rb/Cs})\text{Pb}(\text{Br}_x/\text{Cl}_{1-x})_3$	487	18748	13.6	N/A	21
$(\text{FA/Cs/Rb})(\text{Pb/Sr})(\text{Br/Cl})_3$	487	5700	23.3	42 min @ 50 cd m ⁻²	29
$(\text{PEA/p-F-PEA})_2(\text{K/Cs})_{n-1}\text{Pb}_n(\text{Br}_x/\text{Cl}_{1-x})_{3n+1}$	488	3321	9.56	175 min @ 100 cd m ⁻²	30
$\text{PEA}_2(\text{FA/Cs})\text{Pb}(\text{Br}_x/\text{Cl}_{1-x})_3$	490	~1500	25.87	720 s @ 1 mA cm ⁻²	31
$(\text{FA/Cs})\text{Pb}(\text{Br}_x/\text{Cl}_{1-x})_3$	502	7449	21.5	318.8 min @ 1000 cd m ⁻²	32
$(\text{MOPA/PBA})_2\text{Cs}_{n-1}\text{Pb}_n\text{Br}_{3n+1}$	487	2545	25.4	845 min @ 100 cd m⁻²	Our work

Supplementary References

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